**SUB: Amity JAS25 MCA Cyber Sem2 Security Threat and Hunting**

# Incident Response Assignment (Splunk)

***Attack Type: Brute Force Attack***

**ACADEMIC YEAR**

**2025-2026**

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## TOOL & FRAMEWORK USED

- Splunk (SIEM – Security Information and Event Management)
- NIST Incident Response Framework
- SANS Institute Incident Handling Model
- Splunk Enterprise
- Splunk Universal Forwarder
- Windows Server (VMware), Windows Security Logs
- Firewall
- Active Directory/User Accounts

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# ***Introduction to Incident Response***

Incident Response is a **planned and organized method** used by organizations to **handle cyber security incidents**.
A cyber incident means **any activity that can harm systems, data, or users**, such as hacking, malware, phishing, or unauthorized access.

In real life, companies cannot stop all attacks. So instead of only prevention, they prepare a **response plan**.
This plan tells the security team **what to do**, **when to do**, and **how to do** when an attack happens.

**Why Incident Response is Important**

Without incident response:

- Attacks remain unnoticed
- Hackers stay longer inside systems
- Data gets stolen
- Business operations stop

With incident response:

- Attacks are detected early
- Damage is controlled
- Systems are restored safely
- Future attacks are prevented

Incident Response works like **emergency services** for cyber security.

## ***2. What is a Brute Force Attack***

A **Brute Force Attack** is a cyber attack where an attacker **tries many password combinations again and again** until login is successful.

The attacker does not try to be smart; instead, they try to be **persistent and automated**.

**How Brute Force Attack Works**

1. Attacker selects a user account
2. Attacker uses a tool or script
3. Tool tries hundreds or thousands of passwords
4. If password is weak → login succeeds
5. Attacker gets access

**Why Attackers Use Brute Force**

- Easy to perform
- No advanced hacking skills needed
- Works on weak passwords
- Automated tools available freely

**Common Tools Used**

- Hydra
- Medusa
- Ncrack
- Custom scripts

## ***3. Why We Use Splunk for Incident Response***

Splunk is a **Security Information and Event Management (SIEM) tool**.

It is used to:

- Collect logs
- Store logs
- Search logs
- Analyze logs
- Create alerts
- Build dashboards

**Why Logs Are Important**

Logs are **digital footprints** of everything happening in a system.

Example:

- Who logged in
- From where
- When
- Success or failure

Without logs, **no investigation is possible**.

Splunk makes logs **readable and searchable**.

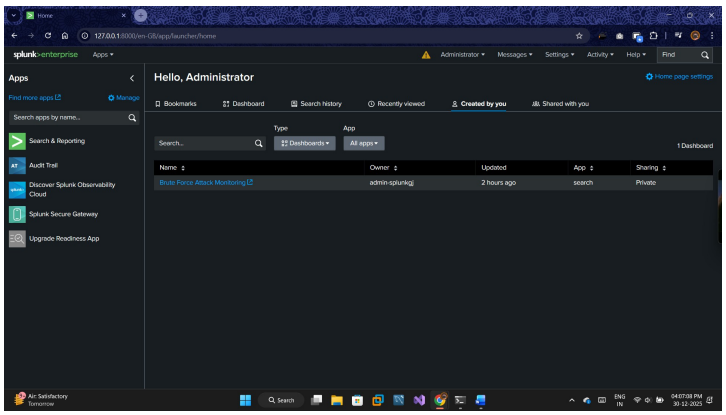
# ***4. Environment Setup***

**Components Used**

1. **Windows Server (VMware)**
   - Acts as the victim system
   - Generates login logs
2. **Splunk Universal Forwarder**
   - Installed on Windows Server
   - Collects logs
   - Sends logs to Splunk Enterprise
3. **Splunk Enterprise**
   - Installed on main system
   - Receives logs
   - Analyzes data

**Why Two Systems?**

- Real-world companies collect logs from **multiple machines**
- Forwarder simulates real enterprise environment
- Separation improves security and performance



0.1- Splunk Enterprise on Host System and Windows Server VM Environment

This screenshot shows the complete lab environment. Splunk Enterprise is installed on the host system, while Windows Server is running inside VMware. The Windows Server acts as a log source, and Splunk Enterprise acts as the log analysis and monitoring tool.

# 5. Installing Splunk Universal Forwarder

The Splunk Universal Forwarder is a **lightweight agent**.

**Why Forwarder is Needed**

Splunk Enterprise **cannot directly read logs** from another system.
The forwarder acts like a **messenger**.

**What Forwarder Does**

- Reads Windows logs
- Converts them to Splunk format
- Sends them securely to Splunk server

**Installation Steps**

1. Download Universal Forwarder for Windows
2. Run installer as Administrator
3. Accept license
4. Enter:
    - Splunk Server IP
    - Port: 9997
5. Set service to start automatically
6. Finish installation

Now logs can travel from Windows Server → Splunk Enterprise.

# 6. Registering Forwarder as Deployment Client

A deployment client allows **central control** from Splunk Enterprise.

**Why This Step Is Important**

Without registration:

- Manual configuration needed
- Hard to manage multiple forwarders

With registration:

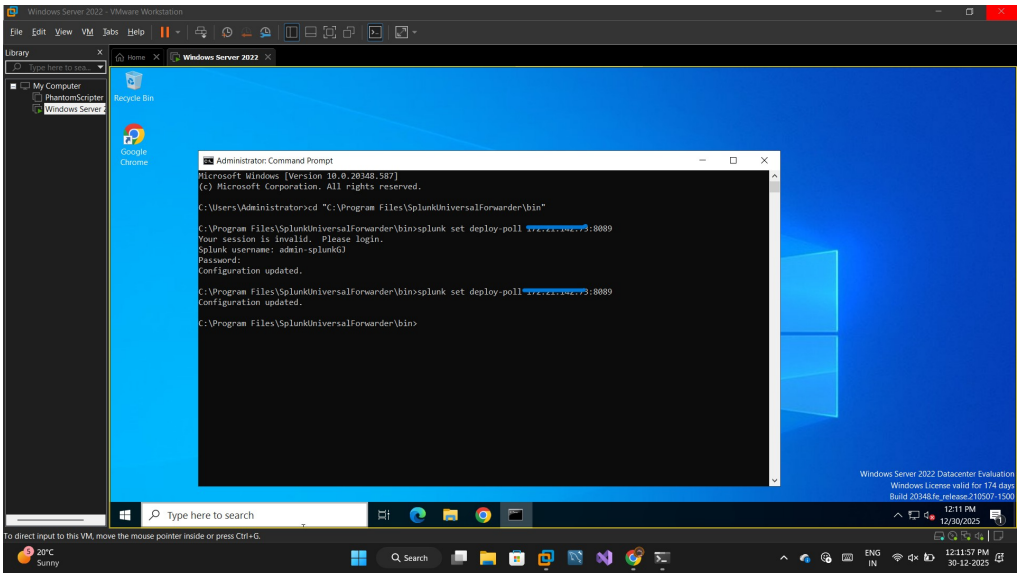
- Central control
- Easy updates
- Policy-based management

**Command Used**

splunk set deploy-poll 168.74.174.73:8089

**What This Command Means**

- deploy-poll → checks for configuration
- 8089 → Splunk management port




02.- Splunk Forwarder Registered as Deployment Client

This screenshot shows that the Splunk Universal Forwarder is successfully registered as a deployment client. This allows centralized configuration and management of the forwarder from Splunk Enterprise.

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## ***7. Enabling Windows Authentication Logs***

By default, **Windows does not log everything**.
For brute force detection, **failed login events must be logged**.
**Why Authentication Logs Are Needed**
Brute force attacks only show up as:

- Repeated failed logins

Without enabling these logs:

- Splunk cannot detect attack
- No evidence is available

**Steps**

1. Open Local Security Policy
2. Go to Audit Policy
3. Enable:
    - Audit Logon Events (Failure)
    - Audit Account Logon Events

Now Windows starts recording login failures.

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## ***8. Configuring Log Source in Splunk***

Splunk must be told **which logs to collect**.

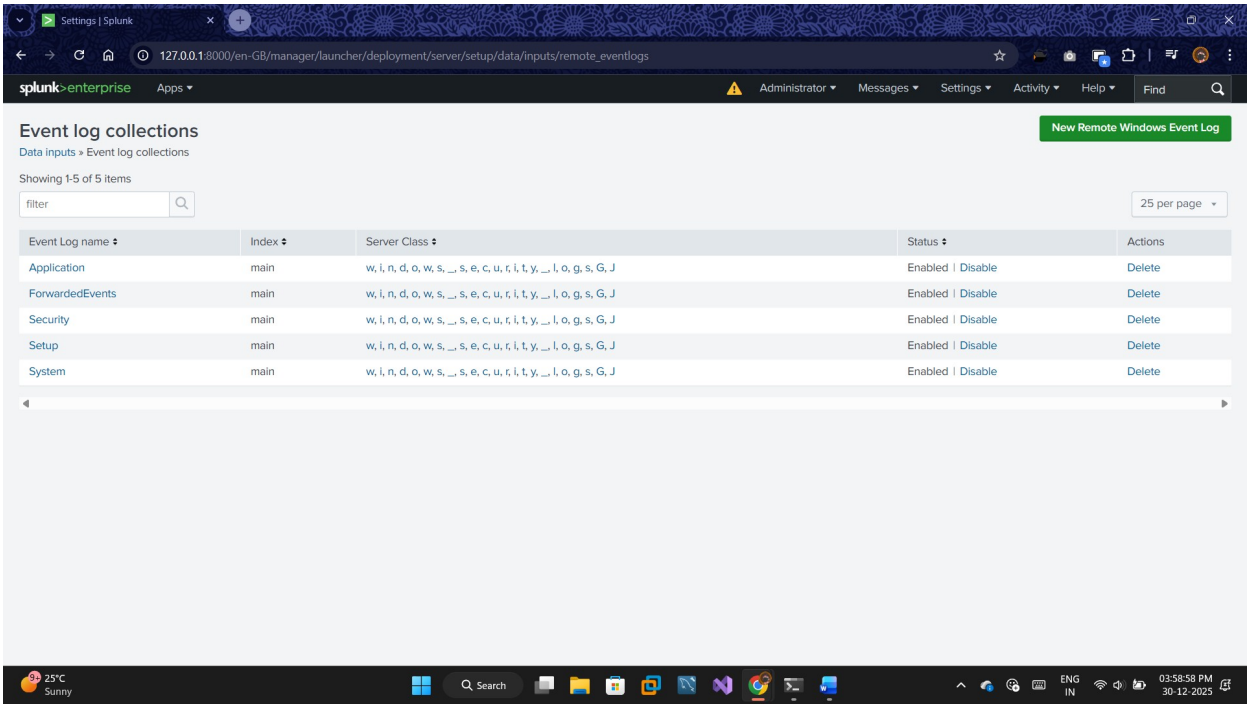
**Why Configuration Is Needed**

- Forwarder collects many logs
- Splunk must know **which ones matter**

**Steps**

1. Splunk Web → Settings → Data Inputs
2. Select Windows Event Logs
3. Enable:
    - Security (MOST IMPORTANT)
4. Choose index = main
5. Save

Now Splunk starts receiving authentication logs.




04.- Configuring Windows Server as Log Source in Splunk

In this step, Windows Server is configured as a log source in Splunk. Splunk Enterprise is set to receive data on port 9997, while the Splunk Universal Forwarder installed on Windows Server sends Windows Event Logs to Splunk. This configuration ensures that security-related events such as authentication logs are successfully collected and indexed for monitoring and analysis.

## ***9. Understanding Event ID 4625***

**What Is Event ID 4625**

- Event ID 4625 = **Failed login attempt**
- Generated every time login fails

**Why This Event Is Key**

- Brute force = many failed logins
- This event proves attack attempt

**Important Fields**

- Account_Name
- Source IP
- Logon Type
- Time

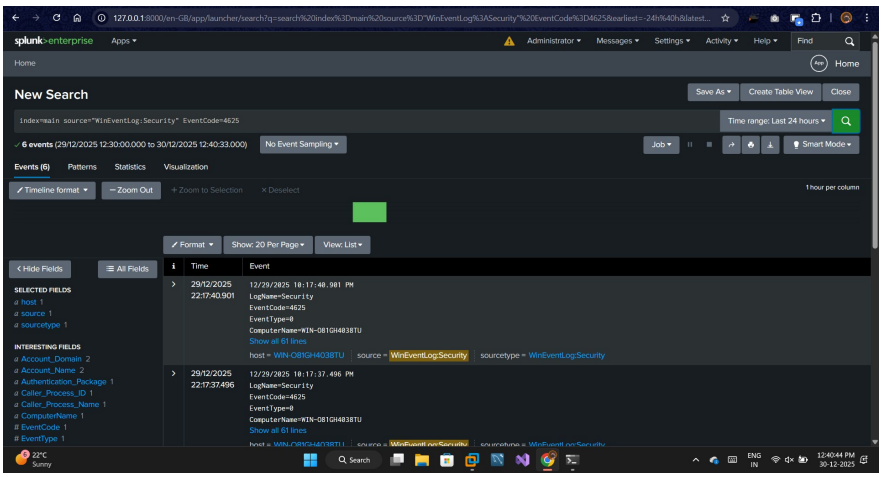
## ***10. Splunk Search Query***

```
source="WinEventLog:Security" EventCode=4625
```

**Explanation**

- source → where log came from
- Security → Windows security logs
- EventCode=4625 → failed login

This search filters **only brute force related events**.

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## ***11. Detecting Brute Force Attack***

```
source="WinEventLog: Security" EventCode=4625 | stats count by Account_Name
```

**Why stats is Used**

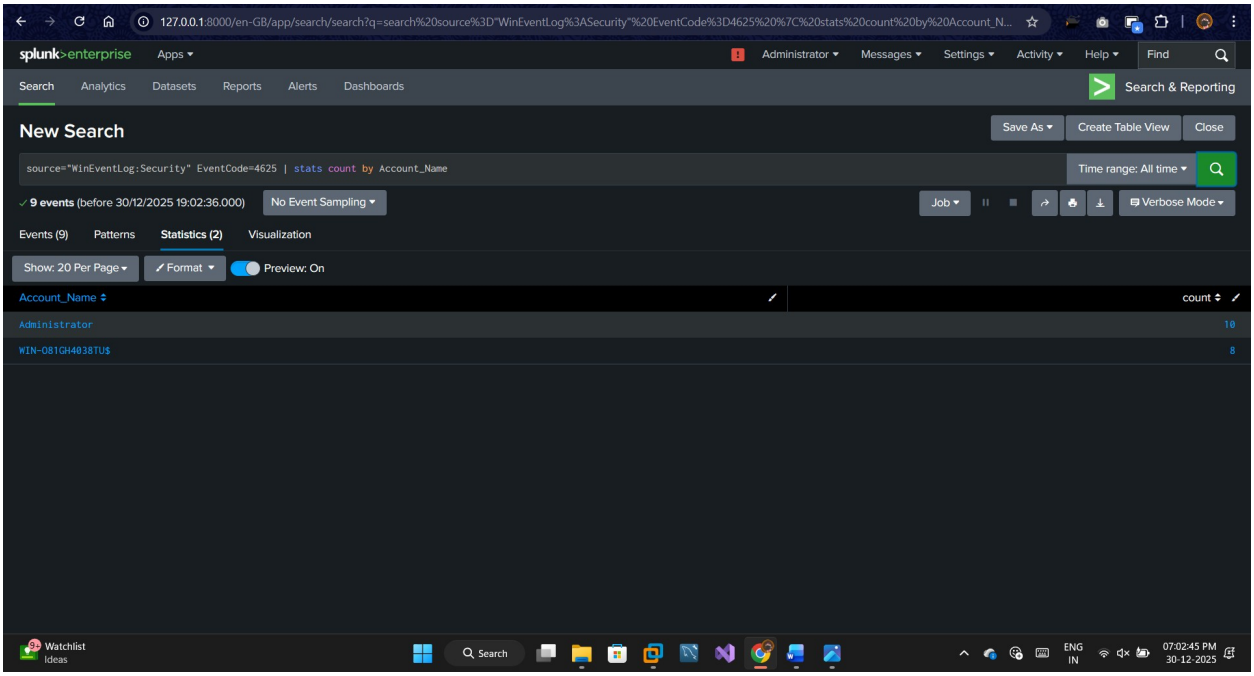
- Converts raw logs into numbers
- Helps identify attack intensity

**What Output Shows**

- Username

- Number of failed attempts

High count = attack in progress.




05.- Detection of Brute Force Attack Using Failed Login Events (Event ID 4625)

In this step, a brute force attack is detected using Windows failed authentication logs. Event ID 4625 represents failed login attempts. By analyzing multiple failed login events for the same user account within a short time, Splunk helps identify brute force attack behavior. This detection allows the security team to take immediate incident response actions.

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## ***12. Visualization Explained***

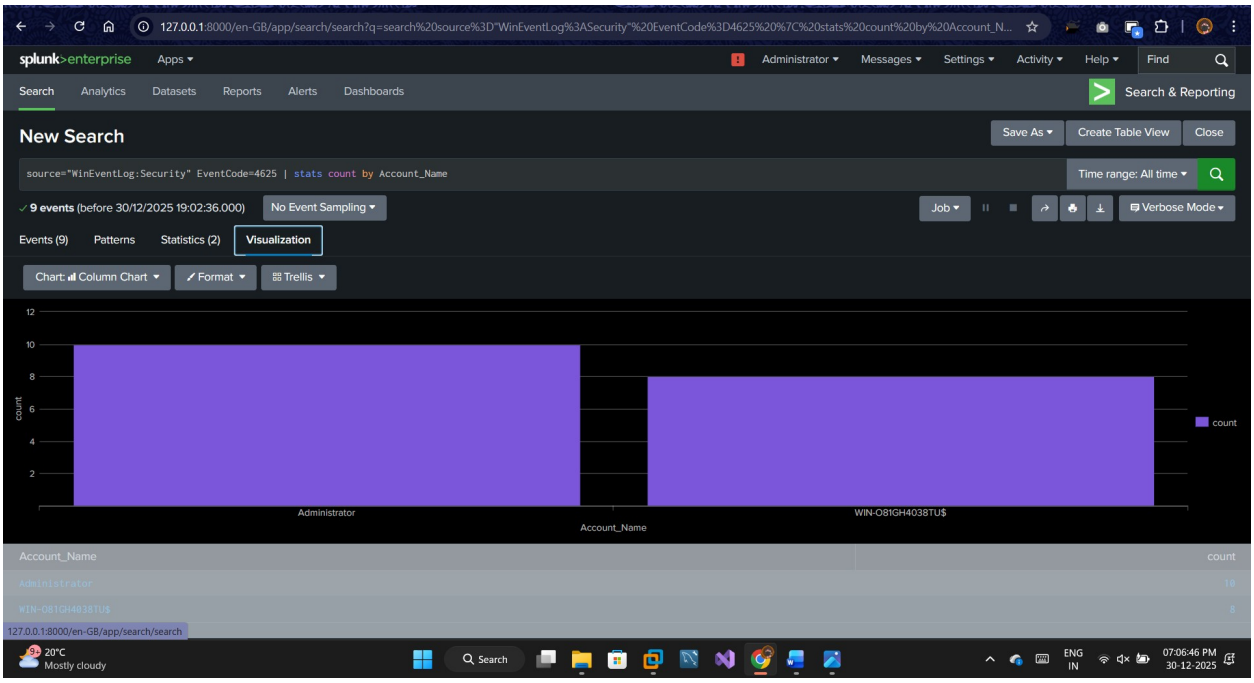
Visualization converts data into **human-readable graphs**.

**Why Visualization Is Needed**

- Security team can understand faster
- Management can see attack severity

**Column Chart Meaning**

- X-axis → User account
- Y-axis → Failed login count



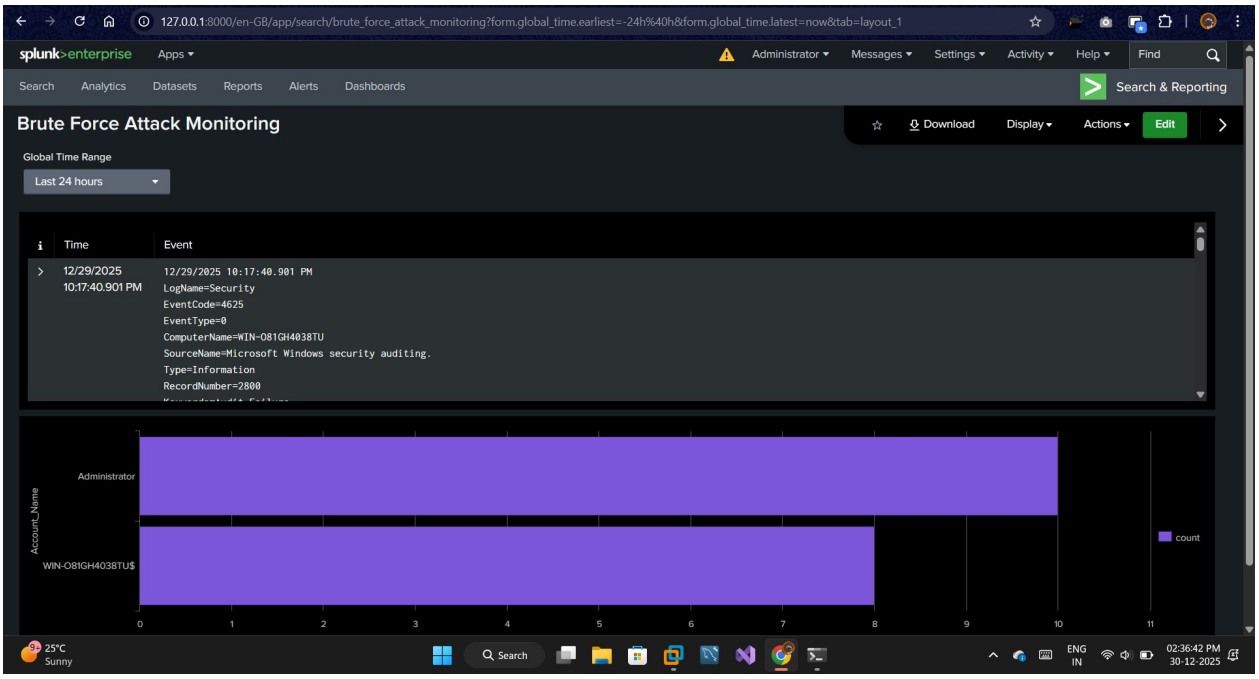

## *13. Dashboard Creation*

Dashboard shows:

- Live attack data
- No need to run search again

**Why Dashboards Are Important**

- Continuous monitoring
- Central view
- Faster decisions




**Brute Force Attack Monitoring Dashboard in Splunk**

In this step, a dashboard is created in Splunk to monitor brute force attacks. The dashboard displays a column chart showing the number of failed login attempts for each user account using Windows Event ID 4625. This dashboard provides a centralized view for security analysts to quickly identify suspicious login activity and respond to potential brute force attacks efficiently.

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## ***14. Alert Creation***

Alert = **automatic warning system**

**Why Alerts Are Needed**

- Admin cannot watch dashboard 24/7
- Alerts notify automatically

**Trigger Logic**

- If failed login > 0
- Alert fires

- Security team notified

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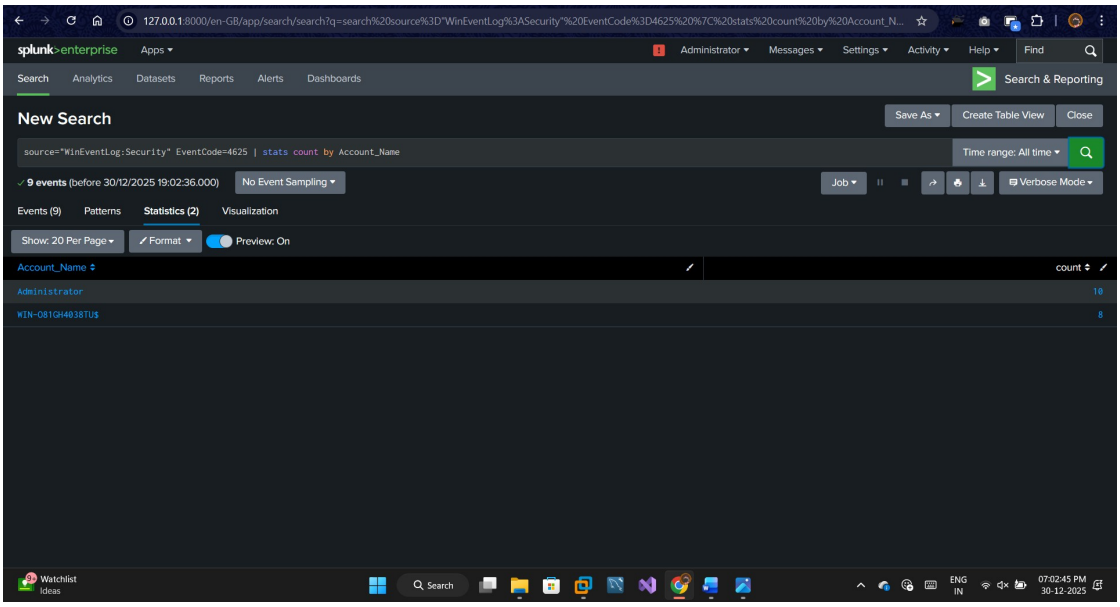
## ***15. Incident Detection Phase***

**Incident Detection** means **finding a security problem** by monitoring logs and alerts.
In this assignment, the incident is a **Brute Force Attack**, detected using **multiple failed login attempts**.

This is the **first IR phase**.

**What Happens**

- Alert fires
- Logs confirm brute force
- Incident is declared




**06.- Incident Detection Phase – Identification of Brute Force Attack Using Event ID 4625**

In the incident detection phase, Splunk is used to monitor Windows authentication logs. Multiple failed login attempts recorded as Event ID 4625 are analyzed to identify suspicious behavior. A high number of failed login attempts for the same user account within a short time indicates a brute force attack. This phase helps the security team detect the incident early and start the incident response process.

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## *16. Containment Phase*

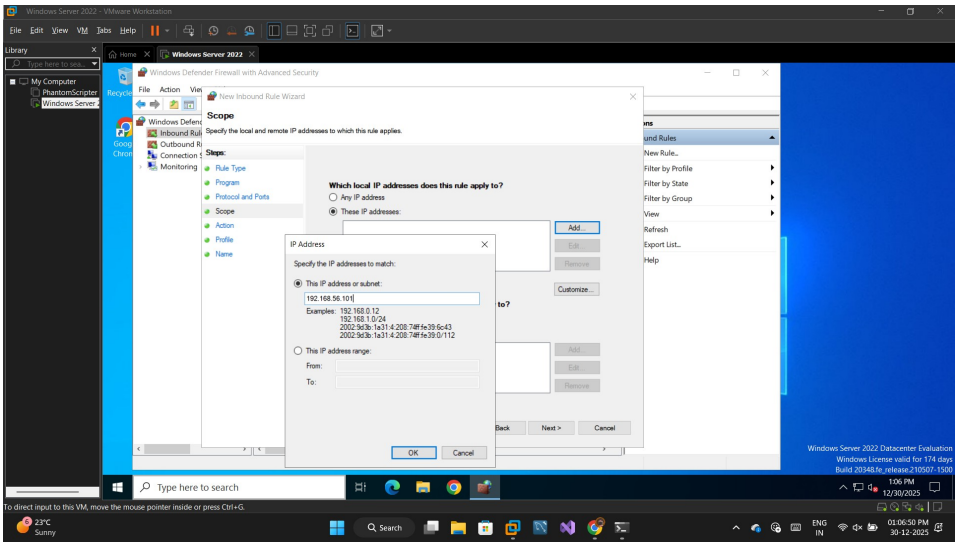
**Why Containment**

- Stop attacker immediately
- Prevent further damage

**Actions Taken**

- Block IP in firewall
- Lock affected account

During the containment phase, immediate action is taken to stop the brute force attack. The affected user account is temporarily locked to prevent further unauthorized login attempts. This action limits the attacker's ability to continue password guessing and protects the system from compromise. Containment helps control the incident before moving to eradication and recovery.




**07.- Firewall Rule Created to Block Attacker IP Address**

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## *17. Eradication Phase*

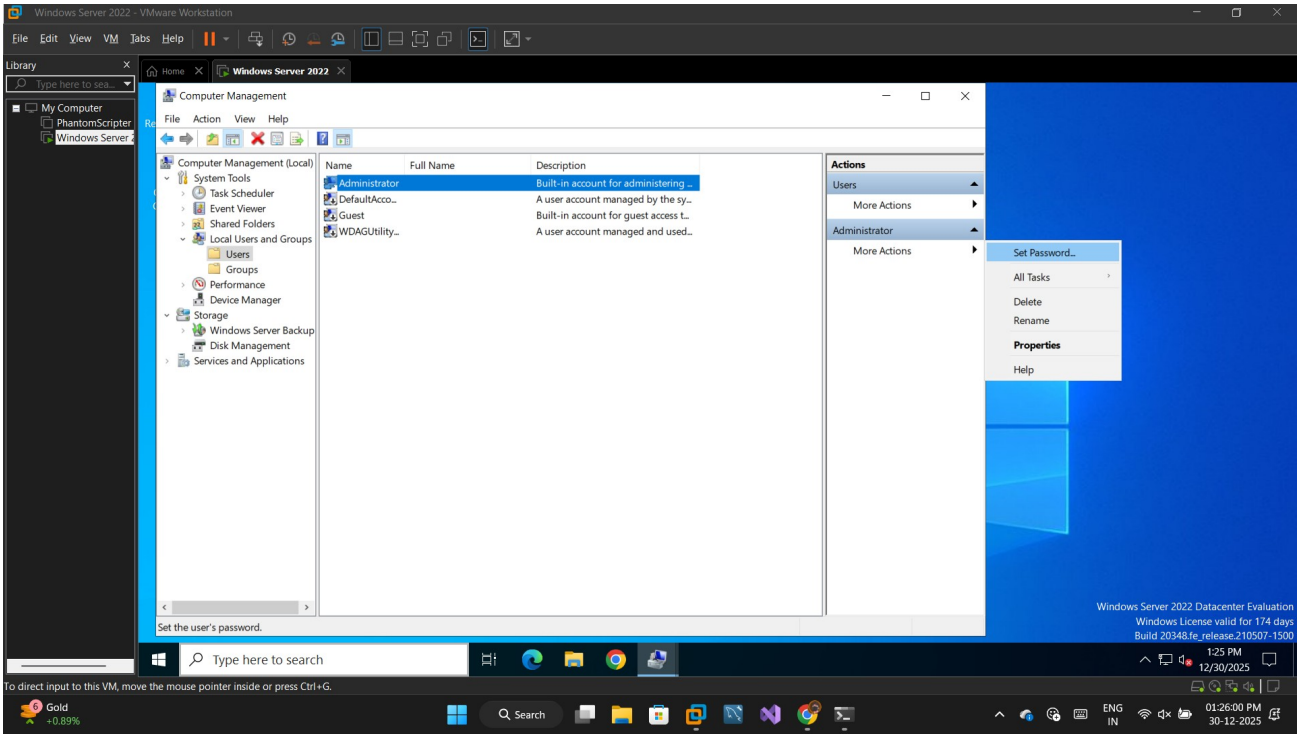
**Why Eradication**

- Remove attack source completely

**Steps**

- Reset password

- Remove malicious IP
- Scan system




### 08.- Incident Eradication by Resetting Compromised User Password

In the eradication phase, the root cause of the brute force attack is removed. The affected user account password is reset to a strong password to prevent further unauthorized access. Additional security policies such as password complexity are reviewed to strengthen system security. Eradication ensures that the attacker can no longer exploit the same weakness.

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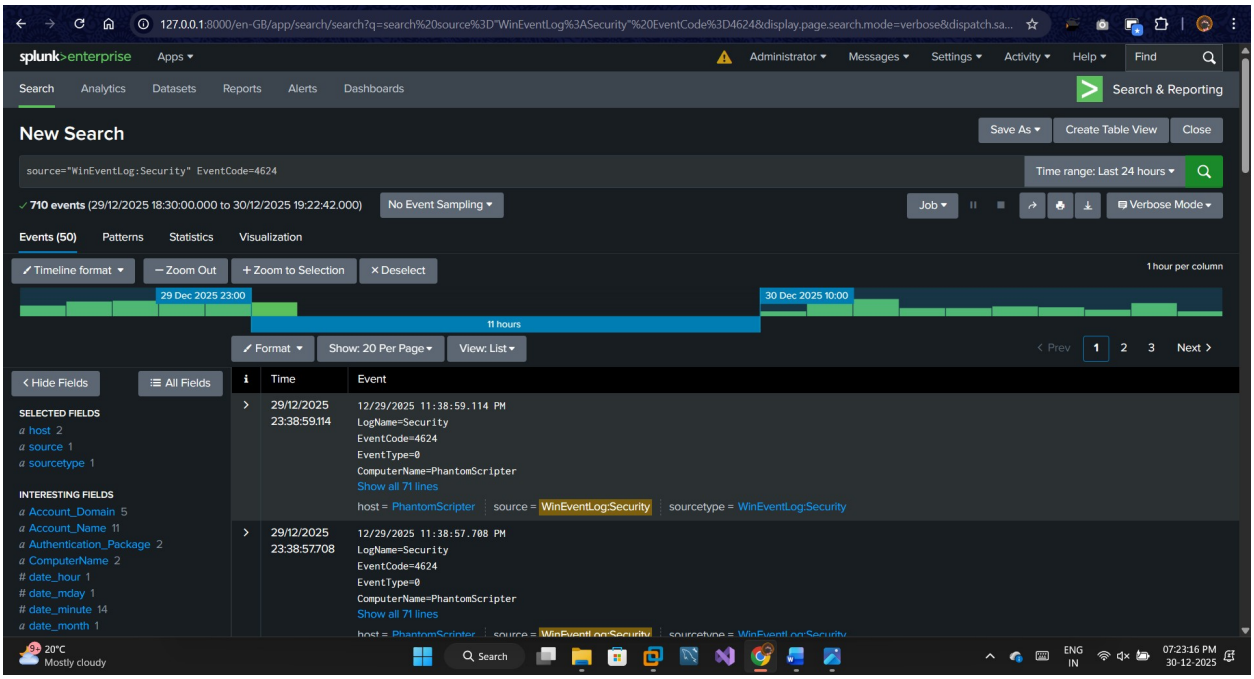
## ***18. Recovery Phase***

### Purpose

- Restore normal operation safely

### Steps

- Unlock account
- Monitor successful logins
- Ensure system stability




### 09- Incident Recovery Phase Showing Successful Login After Remediation

In the recovery phase, the affected user account is restored to normal operation after the brute force attack has been contained and eradicated. The account is unlocked and a successful login is verified using Windows Event ID 4624. Continuous monitoring in Splunk ensures that the system is stable and no further suspicious activity is observed.

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## ***19. Post-Incident Activity (Lessons Learned)***

### Why This Phase Matters

- Improve future security
- Fix weak points

### Learnings

- Weak passwords invite attacks
- Alerts save time

- Monitoring is essential

In the post-incident activity phase, the brute force attack is reviewed to understand how it occurred and how it was handled. Security controls and alert thresholds are improved to prevent similar attacks in the future. Lessons learned from the incident help strengthen system security and enhance the overall incident response process.

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## ***20. Conclusion***

This assignment demonstrated a **complete incident response lifecycle** using Splunk.
A brute force attack was detected using Windows Security logs and Event ID 4625.
Splunk dashboards and alerts helped in real-time detection.
Incident response steps ensured quick containment, eradication, and recovery.

This practical approach reflects **real-world SOC operations**.